

Kishan Narsingh Chhari

Embedded Systems & Robotics Engineer · Real-Time Firmware | Autonomy | Applied AI

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Work eligibility: France — post-study work permit / Passeport Talent eligible



Embedded systems and robotics engineer (M.Sc., ESIGELEC) who builds full systems end-to-end — from real-time firmware and custom PCBs to autonomous control and on-device AI. Ranked 1st in cohort in C Programming and Real-Time Operating Systems; holder of a granted patent and two international publications. Currently researching Vision-Language-Action models for robotic manipulation on a 7-DOF arm.

CORE SKILLS

Languages	C, C++, Embedded C, Python, MicroPython, MATLAB
Embedded & RTOS	STM32 / ARM Cortex-M, ESP32, Raspberry Pi, BeagleBone, MSP430 · FreeRTOS, μ C/OS-II, Embedded Linux · bare-metal, drivers, System-on-Chip
Robotics & Control	ROS / ROS2, RoboDK · PID & LQR control, sensor fusion (Kalman / complementary), inverse kinematics, autonomous navigation
AI & Vision	OpenCV, PyTorch, TensorFlow, TinyML / edge AI, image processing
Protocols & HW	I2C, SPI, UART, CAN, BLE, Zigbee · PCB design (KiCad), SolidWorks, ANSYS, XFLR5, 3D printing
Tools	Git, Keil μ Vision, Code Composer Studio, Proteus, MATLAB/Simulink, VHDL (basics)

EXPERIENCE

Research Intern (M.Sc.) — Robotic Manipulation & Learning — CESI LINEACT (with IMT Atlantique), Rouen	2026
- VeggieBot project: parametric representation of robotic cooking gestures on a 7-DOF UFactory xArm7 with force/torque sensing. - Implementing and comparing learning-from-demonstration, reinforcement learning, Dynamic Movement Primitives and Vision-Language-Action (VLA) models in simulation and on hardware. - Stack: Python, PyTorch, ROS-based robotic simulation, digital twins; contributing toward a research publication.	
Assistant Technical Manager — Robotics — Learning Links Foundation, Mumbai	Oct 2023 – Aug 2024
- Ran a robotics lab (annual budget ₹1 crore; components >₹50 lakh); grew student project & competition participation by ~200%. - Mentored 4+ national-competition teams (3 award-winning, incl. First Lego League & WRO); built the Arduino / ESP32 microcontroller & electronics curriculum.	
Robotics Trainer — Witty International School, Mumbai	2022 – 2023
Established the school's STEAM / robotics program (~900 students) and guided ~13 hands-on student robotics projects from concept to demonstration.	

KEY PROJECTS

Self-Balancing Autonomous Robot — LQR control (upgraded from PID) with complementary-filter IMU fusion on NodeMCU / MPU9250; custom 2-layer PCB (BTS7960 drivers); Raspberry Pi 4 vision (OpenCV) + LD06 LiDAR obstacle avoidance with grid planning; SPI low-latency control link; GPT-4o Realtime voice commands.
Autonomous Fixed-Wing UAV — Pixhawk / ArduPilot AUTO mode with PID AutoTune, ~2 m navigation accuracy and dual-GPS RTL failsafe; custom XFLR5-designed low-speed airfoil (published); 50-min endurance, 2 kg payload, 80 km/h cruise, 3 kg MTOW.
3D-Printed 4-DOF Robotic Arm — SolidWorks design with custom gears and bearing housings; inverse-kinematics joint control in Python on Raspberry Pi driving steppers; validated in RoboDK; 2 kg payload at ~5 mm positioning accuracy.
AI Voice Assistant & Surveillance Node — Raspberry Pi 4 edge vision with OpenCV; periodic scene analysis via ChatGPT API and real-time Telegram alerts across 10+ locations.

EDUCATION

M.Sc. Electronic Embedded Systems	2024–2026
ESIGELEC, Rouen · Embedded Linux, RTOS, Robotics, SoC	
1st in cohort: C Programming (20/20), Real-Time OS (19.05/20), R&D Project (20/20); System-on-Chip 19.5/20	
B.Tech. Mechanical Engineering	2018–2022
Lovely Professional University · Robotics & Automation	

PUBLICATIONS & PATENT

Patent — "Bionic Arm", IP India, 2023 (No. 202311042415)
Aerodynamic & Bending Analysis of Low-Speed Airfoils — <i>Materials Today: Proceedings</i> , 2022
Design & Fabrication of Low-Cost Prosthetic Arm — <i>AIP Conference Proceedings</i> , 2022

AWARDS & CERTIFICATIONS

1st place — Robotics / UAV / CAD at IIT Bombay, IIT Ropar & LNMIIT · 3rd — TECHFEST, IIT Bombay · Certified SolidWorks Associate (CSWA) · Sensor Manufacturing & Process Control (CU Boulder)

LANGUAGES

English — Full Professional · Hindi / Marathi — Native · French — B1 (improving toward B2)
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